

Recovering and Redirecting Peptide Backbone Conformation via Contact-Conditioned C α Generation in a Cross-Reactive pMHC–TCR System

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Abstract

Cross-reactive TCR–pMHC systems, in which one receptor engages chemically distinct peptides through distinct bound backbone conformations, raise a question orthogonal to conventional epitope design: can a structure-conditioned generative model be steered, from receptor-side information alone, toward a specific target backbone conformation — including, deliberately, one other than the conformation nearest its conditioning input? Using a minimal two-state system — the DMF5 TCR bound to HLA-A*02:01, which accommodates two different decameric peptides (GIG and DRG) via backbones separated by 2.87Å C α RMSD — we ask whether RFdiffusion, conditioned only on receptor-side contact residues and given no peptide-backbone template, can generate a C α backbone that recovers a peptide’s own native conformation or crosses into the alternate one. We score *de novo* backbones against a three-criterion test calibrated to native molecular-dynamics ensembles rather than to a single crystal structure, and address a recurring evaluation pitfall by correcting for a backbone-threading artifact that raw RMSD cannot detect. We find that both recovery and redirection are achievable but rare, gated by the breadth of conditioning rather than by any specific curated residue set, and that once that breadth is met their frequency is governed by sampling depth. A templating control further shows that directly supplying backbone geometry constrains the target conformation far more tightly than contact conditioning alone, locating register in a spatially confined signal that receptor-side contacts only weakly determine. These results, from a single system, map the current capability boundary of contact-conditioned backbone generation and separate what it can retrieve from what must still be supplied as explicit geometry.

1 Introduction

Cross-reactive TCR–pMHC pairs that engage chemically distinct peptides via distinct bound backbone conformations pose a design question orthogonal to conventional epitope design: can a structure-generative model be conditioned, using only information available on the receptor side, to produce a specific target backbone conformation — including, deliberately, the conformation other than the one nearest to the conditioning input? We address this question exclusively

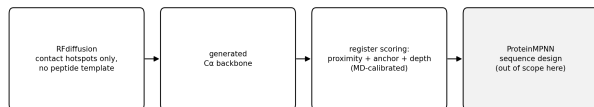


Figure 1: **Pipeline schematic.** RFdiffusion (contact-hotspot conditioning, contig 10–10, no peptide template) → generated C α backbone → three-criterion register scoring against MD-calibrated thresholds → (downstream, out of scope here) ProteinMPNN sequence design.

at the C α backbone level. RFdiffusion generates backbone coordinates without an associated sequence at this stage of the pipeline; consequently every metric defined here operates on C α geometry, treated throughout as a geometric rather than sequence property. GIG and DRG are not the same peptide sequence adopting two registers of a shared groove placement — they are two different decamers (SML-GIGIVPV and MMWDRGLGMM) that DMF5 happens to recognize via two different bound backbone shapes. We use “register” only as shorthand for “which of the two backbone conformations,” never to imply a shared sequence.

A central methodological problem in evaluating this kind of claim is that “resembles the native peptide” has historically been operationalized as C α RMSD to a single crystal structure, an unscaled quantity: a design at 1.2Å cannot be judged “inside” or “outside” a conformational envelope without knowing how wide that envelope is. We address this by calibrating acceptance thresholds against a native 310K/50ns MD ensemble of both conformations (Methods §2.4), and by requiring three independent criteria to agree (proximity, anchor identity, burial depth) rather than relying on proximity alone, which is separately confounded by a backbone-threading artifact (§3.4) that a raw-RMSD-only criterion cannot detect.

We report five results, ordered to lead with the central empirical finding (recovery and crossing both occur and are quantifiable) before addressing what gates their occurrence, what does not explain their rate, a methodological artifact that must be corrected for, and the underlying mechanism.

2 Methods

2.1 Structural system and conformational-state definition

The system is HLA-A*02:01 ($\alpha 1/\alpha 2/\alpha 3$, chain A) with $\beta 2$ -microglobulin (chain B), a 10-residue peptide (chain C), and the DMF5 TCR α/β V-domains (chains D/E), crystallized in two forms: 6AM5, bearing the peptide SMLGI-GIVPV (“GIG”), and 6AMU, bearing the peptide MMW-DRGLGMM (“DRG”) — two different decamer sequences, not one sequence in two groove registers. Component-wise C α RMSD between the two crystals (computed after superposing the rigid MHC $\alpha 1/\alpha 2$ platform) localizes essentially the entire 2.87Å difference between the two complexes to the peptide backbone; the MHC groove floor (0.20–0.39Å), CDR3 α (0.59Å), and CDR3 β (0.47Å) are rigid between the two forms. “Register” is used only as a name for a peptide’s bound C α conformation in a common reference frame (§2.3); it carries no sequence-identity meaning here.

2.2 Backbone generation

De novo peptide backbones were generated with RFdiffusion (Complex_base_ckpt.pt, SE3nv environment), contig specification A1-180/0 B1-100/0 D1-115/0 E1-120/0 10-10 (MHC/B2M/TCR α /TCR β held fixed as structural context; the terminal 10-10 block is fully de novo with no peptide-backbone residues templated), diffuser.T=30 (full reverse-diffusion trajectory) throughout. Conditioning was varied solely via ppi.hotspot_res, ranging from 3 to 42 receptor-side residues across 55 (crystal $\times$ conditioning) cells drawn from 5 campaigns (internal names: grind, ladder, promising, denovo30, maxcond), totaling 1,295 scoreable 10-residue designs after filtering (§2.3). We name each conditioning cell by its hotspot count (e.g. C18 = 18 receptor-side hotspot residues) rather than by internal campaign codename, since contact count is the property under discussion throughout §3; where the same count was tested on both crystals we disambiguate with a crystal suffix, e.g. C24 (6AM5) vs. C24 (6AMU). This is the corpus used for the register test in §3.1–3.4; it is distinct from (i) the motif-templating ladder (§3.5), a separate small set of cells (fixall $\rightarrow$ fix0, rfd_recover campaign, $N = 10$ designs/crystal/level) that still carry the full rich hotspot list but vary how much of the native peptide backbone is additionally supplied as a template, and (ii) the null-baseline campaign (§3.6), a new set of cells (null0) carrying no hotspots and no template at all. Of the 55 conditioning cells, 16 pool draws from more than one campaign launched at different times under the same (crystal, conditioning) specification — e.g., the C3–C12 contact-count ladder draws from denovo30, ladder, and promising jointly, and mhc/mhc_tcr2/tcr2 draw from denovo30 and grind jointly. We disclose this because it means the 55 cells are not 55 single-shot, mutually independent draws; they are 55 conditioning *specifications*, several of which were incrementally added to across sequential campaign launches. This is consistent with — and does not undermine — the §3.3 finding that per-cell hit count tracks cumulative draw

count, since draws accumulated for the same cell over time are exactly the repeated-sampling process that finding describes.

2.3 Register scoring

For each design, the MHC $\alpha 1/\alpha 2$ C α trace is superposed onto a common reference frame via robust (outlier-trimmed) Kabsch alignment; the peptide C α coordinates are transformed into that same frame, making them directly comparable to the two native reference conformations (GIG, DRG; each a fixed (10,3) C α array in the same frame) and to a native MD trajectory processed identically.

A design is scored on three criteria, each thresholded against a native MD ensemble of the *target* conformation (own conformation for “recovery,” alternate conformation for “crossing”):

1. **Proximity** — mean C α RMSD to the target’s 10-residue reference $\leq$ (native MD mean + 3σ) for that conformation.
2. **Anchor identity** — the design’s C α position with minimum distance to a fixed F-pocket centroid (defined from six HLA-A*02 F-pocket residues) must equal the target’s native anchor position (P10 for GIG, P9 for DRG).
3. **Burial depth** — that minimum F-pocket distance must fall within (native MD mean $\pm 3\sigma$) for the target conformation.

All three must hold simultaneously. This retires two previously used but unreliable proxies: raw RMSD-only thresholding (which never registers a crossing under any threshold in the 1.4–2.5Å range tested historically, a false-negative failure mode) and F-pocket-argmin proximity alone without a depth requirement — 388/1,295 designs (30%) satisfy this alone, which we consider false-positive-prone, since it is spoofable by centroid-adjacent scatter unrelated to genuine anchor burial.

Threading classification. Each design’s F-pocket-proximal anchor position was additionally classified as *forward* (positions P7–P10 occupy the F-pocket, matching both native conformations’ actual anchor placement) or *reverse* (P1–P4), with P5–P6 as an ambiguous buffer zone; this classification was derived from the empirical bimodal distribution of anchor position across the full corpus (§3.4) and is applied as a filter before all recovery/crossing statistics reported in §3.1–3.3 (forward-threaded subset, $N = 600$).

2.4 Native MD calibration

Explicit-solvent MD (Amber ff19SB force field, TIP3P water, 12Å periodic box, 0.15M NaCl, 2fs timestep, minimization to convergence, $\leq 10,000$ steps) was run for each crystal structure at 310K (body temperature; 2ns equilibration, 50ns production). Each native conformation is thereby represented by its 1,000-frame thermal ensemble rather than a single crystal structure, and acceptance thresholds are calibrated to that ensemble.

Because the thresholds are estimated from a finite, temporally autocorrelated trajectory, we do not treat them as exact: we quantify their sampling uncertainty with a moving-block

conformation	proximity (95% CI)	F-pocket depth	outcome	count/ N	rate (95% CI)
DRG (6AMU), primary	$\leq 1.48\text{\AA}$ [1.36–1.58]	[3.95, 6.02] $\text{\AA}$	DRG recovery (6AMU), full pool	4/8,722	0.046% (0.01–0.12%)
GIG (6AM5), under-sampled	$\leq 1.65\text{\AA}$ [1.30–1.68]	[3.48, 4.79] $\text{\AA}$	DRG crossing (6AM5), full pool	3/8,722	0.034% (0.01–0.10%)
			DRG recovery, fwd. subset	4/4,214	0.095% (0.03–0.24%)
			DRG crossing, fwd. subset	3/4,214	0.071% (0.02–0.21%)

Table 1: Calibrated acceptance bands (310K/50ns MD) with block-bootstrap 95% CIs. A design whose $C\alpha$ RMSD falls within a band’s CI is reported as consistent with the native envelope within calibration uncertainty. The GIG band is under-sampled ($ESS \approx 2$) and is not used to adjudicate redirection.

bootstrap (block length twice the integrated autocorrelation time, 4,000 resamples), reporting each threshold as a point estimate with a 95% confidence interval. Convergence is uneven between the two runs — the DRG (6AMU) trajectory is well sampled (integrated autocorrelation time 1.9ns; effective sample size ≈ 13 of 1,000 frames) whereas the GIG (6AM5) run is not (≈ 2) — so we treat the DRG band as the primary calibration and present register *redirection* exclusively as recovery of the DRG conformation from a 6AM5 scaffold, adjudicated against the well-converged DRG band. The resulting bands are:

The two native conformations remain cleanly separated basins — each native ensemble sits $\sim 0.9\text{\AA}$ from its own reference, far inside the $2.87\text{\AA}$ crystal-to-crystal separation — so the criteria discriminate register rather than merely detecting a groove-resident backbone. As a negative control, unconditioned draws (no hotspots, no template) never enter either acceptance envelope.

2.5 Null/background campaign

To measure the chance rate at which RFdiffusion produces a backbone landing inside either acceptance envelope with **no** conditioning information at all, we generated a separate set of designs (`null0` cells, one per crystal) using the same contig (10–10, no template) but omitting `ppi.hotspot_res` entirely — the receptor chains are still held fixed as structural context, but no contact information is supplied. This is distinct from the `fix0` cells used in the templating ladder (§3.5), which — despite the name — still carry the full rich hotspot list; `fix0` varies template fraction at fixed (maximal) hotspot conditioning, whereas `null0` removes hotspot conditioning entirely. Status and current N are reported in §3.6.

2.6 Statistical analysis

Proportions are reported with Clopper–Pearson exact 95% confidence intervals. Comparisons between conditioning tiers use Fisher’s exact test; where one cell of the 2×2 table is zero, we additionally report a Haldane–Anscombe continuity-corrected odds ratio (+0.5 to all cells) with its associated 95% CI. Where multiple (55) conditioning cells were compared to identify an outlier cell, the resulting p -value is Bonferroni-corrected for 55 comparisons. The relationship between per-cell sample size and per-cell hit count was assessed by Pearson correlation across all 55 cells.

Table 2: Recovery and crossing rates, adjudicated on the DRG register (the well-converged 310K band, §2.4), full de novo pool ($N = 8,722$) and forward-threaded subset ($N = 4,214$; §2.3, §3.4), counting designs within the band’s 95% CI, with Clopper–Pearson exact 95% CIs. Unconditioned null control: 0/932.

3 Results

We generated 8,722 de novo peptide backbones across 20 conditioning schemes (both crystals), together with 932 unconditioned (null) draws and 3,060 templating controls, and scored each against the three-criterion register test described in §2.3, adjudicated against the well-converged 310K DRG acceptance band (§2.4). Several designs entered the native envelope: some recovered their own native conformation, and others matched the *alternate* conformation better than their own. Reaching a clean count of these events, however, first required resolving a structural ambiguity present in every conditioning scheme tested — described in §3.4 — which we identified during this analysis and corrected for before computing any of the statistics below. We present the headline result first (§3.1), then work outward to what explains its pattern (§3.2–3.3), the correction that made it trustworthy (§3.4), and the mechanism underneath (§3.5), closing with the design and status of a direct null-baseline measurement (§3.6).

3.1 Both recovery and crossing occur at low, bounded rates

All three criteria (proximity, anchor identity, depth) were satisfied simultaneously by each reported design; none is a marginal call on a single axis. The closest de-novo crossing design ($C\alpha$ RMSD $1.44\text{\AA}$ to the alternate conformation’s mean, correct anchor position, correct depth) is $1.03\text{\AA}$ from a specific frame sampled at 36.2ns into that conformation’s MD trajectory — closer to an observed thermal conformation than to the register’s time-averaged structure.

3.2 Conditioning breadth, not composition, gates the outcome

To isolate conditioning breadth and sampling depth from raw draw volume, §3.2–3.3 draw on a controlled matched-attempt sub-campaign (equal draws per scheme, §5) rather than on the full de novo corpus of §3.1. Having established that recovery and crossing occur at all, the natural next question was whether the qualifying designs were spread evenly across the conditioning cells of that sub-campaign or concentrated in a particular kind of cell. Conditioning hotspot sets ranged from 3 residues (sparse) to 37–42 (rich). We evaluated whether the composition of “rich” cells reflected anything beyond generic contact frequency by computing,

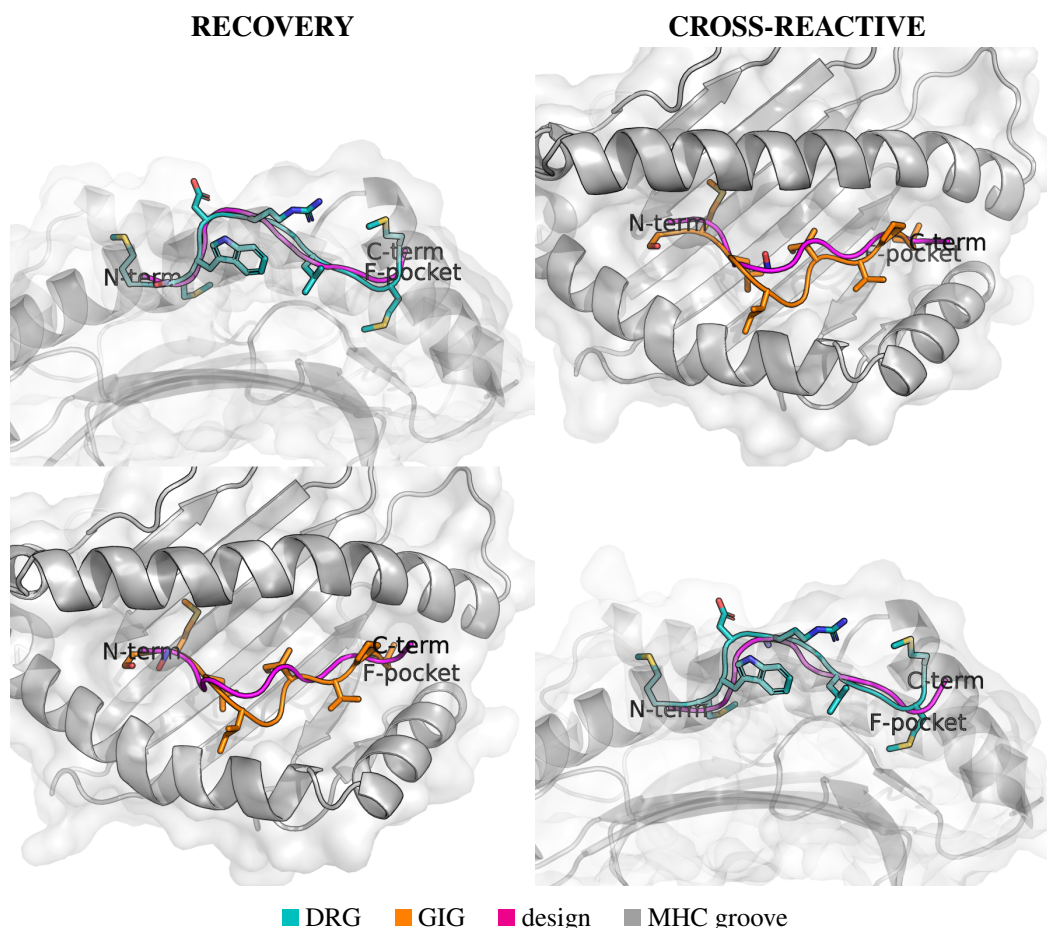


Figure 2: **Recovery and crossing are register-specific, not generic proximity.** Each design is shown against both conformations, establishing that the match to its reported conformation is not simply “close to either.” **Left column (RECOVERY):** the recovering design (magenta) is 1.07Å from its own target DRG (teal, top) but 2.75Å from the non-target GIG (orange, bottom) — inside the 310K DRG acceptance band (§2.4). **Right column (CROSS-REACTIVE):** the closest de-novo crossing design (magenta) is 3.02Å from its own cognate GIG (orange, top) — the conformation it was *not* trying to match — but 1.44Å from the DRG conformation it crossed into (teal, bottom; 1.03Å to the single closest individual MD frame, discussed below), within the DRG band’s 95% confidence interval. Both designs sit substantially closer to the conformation they are credited with matching than to the alternative. Colour key (consistent across all panels): DRG teal, GIG orange, design magenta, MHC groove grey.

directly from the 6AM5 crystal structure, the number of peptide-proximal heavy-atom contacts ($\leq 5\text{\AA}$) per receptor residue:

Every hotspot used in the richer conditioning cells is among the top-ranked residues by this generic, distance-based metric; no residue included at the rich end reflects a curated or mechanistically distinct selection beyond higher raw contact count on both the MHC and TCR side. We therefore treat “rich” as “broad coverage of high-contact-degree residues” rather than as a biologically distinguished category.

Under this operationalization: designs from cells covering ≥ 9 high-contact-degree residues produced 5/718 hits (0.70%, 95% CI 0.23–1.62%); cells limited to the top 3–5 residues produced 0/587 (95% CI 0–0.63%). Fisher’s exact

test: $p = 0.068$ (Haldane–Anscombe-corrected OR= 9.06, 95% CI 0.50–164, reflecting the small event count). This result is suggestive rather than conclusive on its own, but is consistent with conditioning breadth being necessary, even though (§3.3) it is not sufficient to predict which specific rich design succeeds.

3.3 Above the gate, sampling depth explains hit frequency

Conditioning breadth (§3.2) explains why sparse cells never hit, but it does not by itself explain which *rich* cell produces a hit and which does not — the five qualifying designs came from only 3 of the many rich cells tested. To ask whether some rich conditioning schemes are simply better than others, we ran a fixed, matched campaign: 26 designs were gen-

scheme	class	6AM5	6AMU	total	scheme	class	6AM5	6AMU	total
k14	de novo	607	610	1,217	k24	de novo	601	602	1,203
L5_max	de novo	608	608	1,216	max	de novo	601	600	1,201
L3_nterm_t2	de novo	608	606	1,214	fix0	templated	605	607	1,212
L4_expanded	de novo	610	604	1,214	fix6	templated	605	606	1,211
k18	de novo	609	602	1,211	fix2	templated	600	607	1,207
mhc_tcr2	de novo	606	605	1,211	fix4	templated	601	605	1,206
mhc_tcr1	de novo	604	604	1,208	fix8	templated	601	604	1,205
L1_nterm	de novo	601	606	1,207	null0	null	764	769	1,533
L2_nterm_t1	de novo	602	604	1,206	xo_nt5_own	crossover	43	25	68
tcr2	de novo	602	603	1,205	xo_nt4_own	crossover	41	22	63
mhc	de novo	601	602	1,203	xo_nt3_own	crossover	40	21	61
tcr1	de novo	600	603	1,203	xo_nt2_own	crossover	31	21	52
					xo_nt4_cross	crossover	26	21	47
					xo_nt2_cross	crossover	26	21	47

Table 3: Designs generated per conditioning scheme, per crystal (6AM5 = GIG context, 6AMU = DRG context), across the complete converged campaign (24,831 designs total: 16,919 de novo, 6,041 templated, 1,533 null, 338 crossover). The 20 primary schemes were each run to 600 designs per crystal; the crossover schemes are a lower-volume targeted mechanistic probe (§3.6). The register statistics in Tables 2–5 are adjudicated on the frozen de novo analysis pool ($N = 8,722$); this table reports the full generation totals.

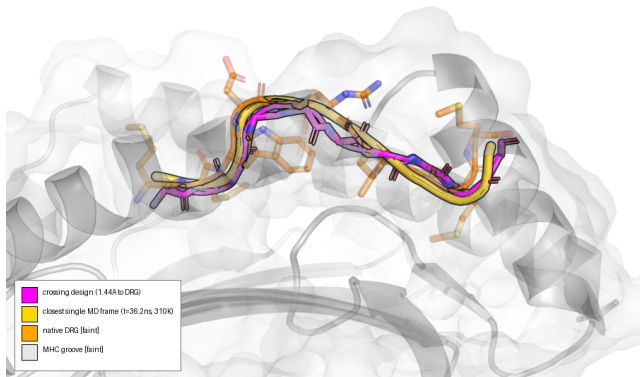


Figure 3: **The closest crossing design against the single closest real MD snapshot.** Magenta: the closest de-novo crossing design (6AM5_L3_nterm_t2, 1.44Å to the DRG register). Gold: the individual 310K/50ns MD frame it is nearest to ($t = 36.2$ ns into the DRG trajectory, 1.03Å away) — an actual sampled thermal conformation, not the register’s time-averaged mean (orange, faint, shown for reference at 1.44Å). The two traces nearly coincide.

erated under each of 8 different rich conditioning schemes (5–42 contacts, both crystals), all drawn before any cross-scheme comparison was made, so that no scheme could be favored by having received more attempts than another:

All eight cells converge to a 0.49Å-wide band regardless of a nearly 9-fold range in contact count. Extending the single best-performing cell (C9) from 26 to 133 draws did not improve on the original best (2.06Å unchanged); the 107

residue	contacts	residue	contacts
A167	10	A70	6
E97 (TCR β)	11	A63	6
A147	9	A67	5
A77	8	A80	5
A159	8	A97	5
A99	8	A143	5
A66	8	D30 (TCR α)	7
A155	7	E96 (TCR β)	4

Table 4: Peptide-proximal heavy-atom contact counts (≤ 5 Å), 6AM5 crystal structure.

conditioning	contacts	best-of-26 (Å)
C9	9	2.06
C18	18	2.06
C5	5	2.24
C42 (6AMU)	42	2.27
C37 (6AM5)	37	2.28
C24 (6AM5)	24	2.40
C18 (6AMU)	18	2.42
C24 (6AMU)	24	2.55

Table 5: Best-of-26 distance to target across 8 matched rich-conditioning cells.

additional draws contributed a worse best (2.36Å) and substantially worse median (13.7Å vs. 4.1Å for the first 26) — consistent with resampling a fixed distribution rather than converging toward the target.

Across all 55 conditioning cells (not just the 8 in the controlled comparison), per-cell hit count correlated with per-cell draw count (Pearson $r = 0.657$, $p < 0.0001$). The single cell with the highest raw hit count (C18, combined across both crystals: $3/186 = 1.61\%$, vs. $2/1,119 = 0.18\%$

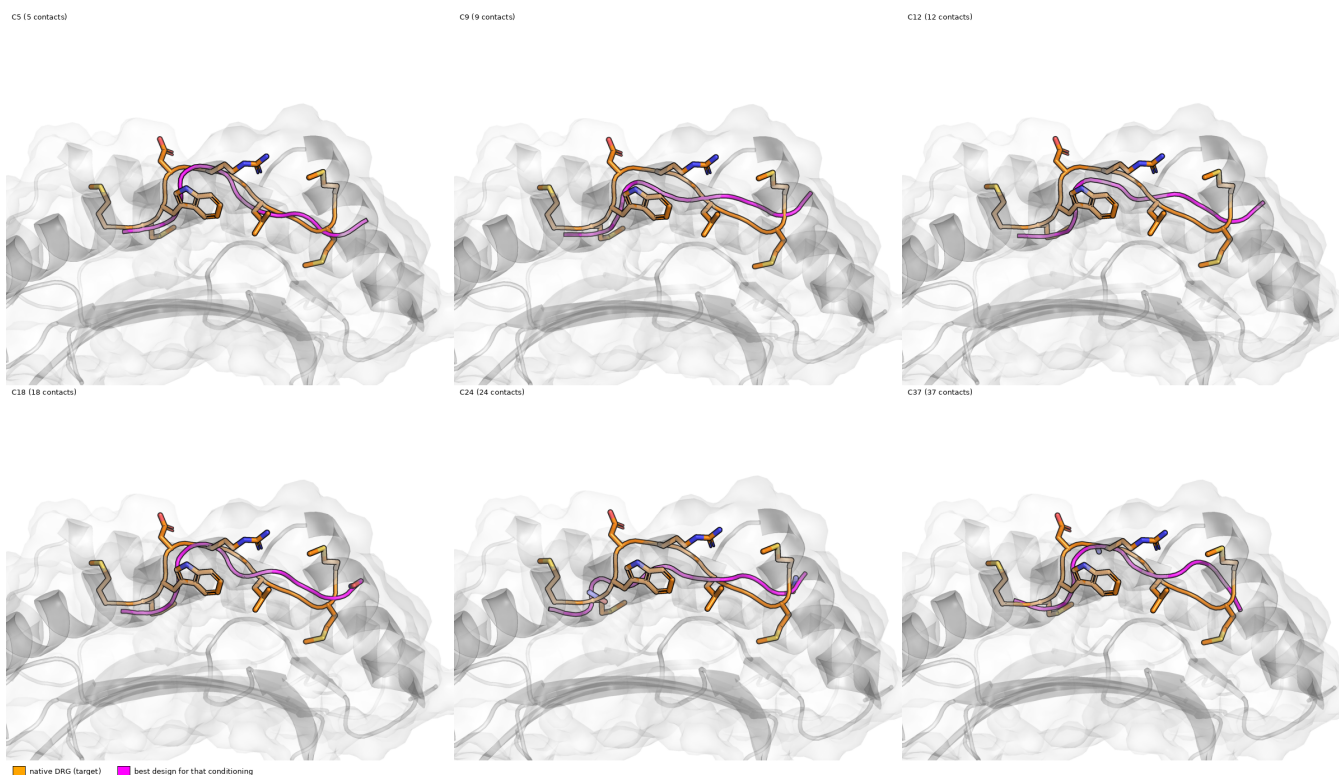


Figure 4: Best design per rich conditioning, one panel each. The best (closest-to-target) design (magenta) found for each of 6 rich conditioning cells (5–37 contacts), each overlaid on the native DRG target (orange) against the MHC groove (white surface). All six land at a similar distance from the target regardless of contact count — the structural counterpart of the 0.49Å-wide best-of-26 band in Table 5. Restricted to the rich tier; combining with sparse-tier cells here would conflate the §3.2 and §3.3 comparisons.

for all other cells combined; Fisher’s exact $p = 0.023$) does not survive Bonferroni correction across the 55 cells examined (corrected $p = 1.0$). We conclude that, conditional on clearing the conditioning-breadth gate identified in §3.2, the identity of the conditioning scheme does not detectably predict outcome; the number of designs drawn does.

A note on directional asymmetry. The hit set contains 4 crossings and 1 recovery — the opposite of what a hypothesis of “conditioning biases toward its own target” would predict. With $n = 5$ total events, a two-sided exact binomial test cannot distinguish this split from 50/50 ($p = 0.375$), so we do not read this as evidence of a directional bias one way or the other.

3.4 A threading artifact affects half of backbones and must be filtered

The results in §3.1–3.3 are already reported net of a correction we describe here. Early in this analysis, per-conditioning distributions of $C\alpha$ RMSD were consistently bimodal — a near-native mode and a second, far mode roughly matched in size — under every conditioning scheme inspected, regardless of contact count. Both native conformations anchor a C-terminal-half residue (P9 or P10) in the MHC F-pocket; tracing the far mode to its cause showed

that, in the absence of any explicit directional constraint, de novo backbones instead anchor their N-terminal residue (P1–P4) in that same pocket in close to half of all draws (4,091 of 8,722 de novo backbones, 46.9%, are reverse-threaded) — a mirror-image backbone, not a genuine register variant, which raw $C\alpha$ RMSD to a crystal structure cannot distinguish from a poorly-folded forward-threaded design. This was quantified in two representative conditioning cells:

Having already found in §3.2–3.3 that conditioning breadth and identity shape whether recovery and crossing occur, we checked directly whether they also shape which way the backbone threads. Across all 36 conditioning cells with a known hotspot count (3–37 contacts, $N = 1,160$ designs), the forward-threaded fraction showed no relationship to contact count (Pearson $r = 0.154$, $p = 0.37$; Spearman $\rho = -0.019$, $p = 0.91$) and stayed within a 40–58% band at every contact-count tier tested. Threading direction is therefore independent of conditioning entirely, not merely similar between the two cells shown above. Per-position deviation profiles for the forward and reverse sub-populations are approximately mirror images of one another; pooling them without this classification produces a spurious “elevated deviation at both termini” signature that is an artifact

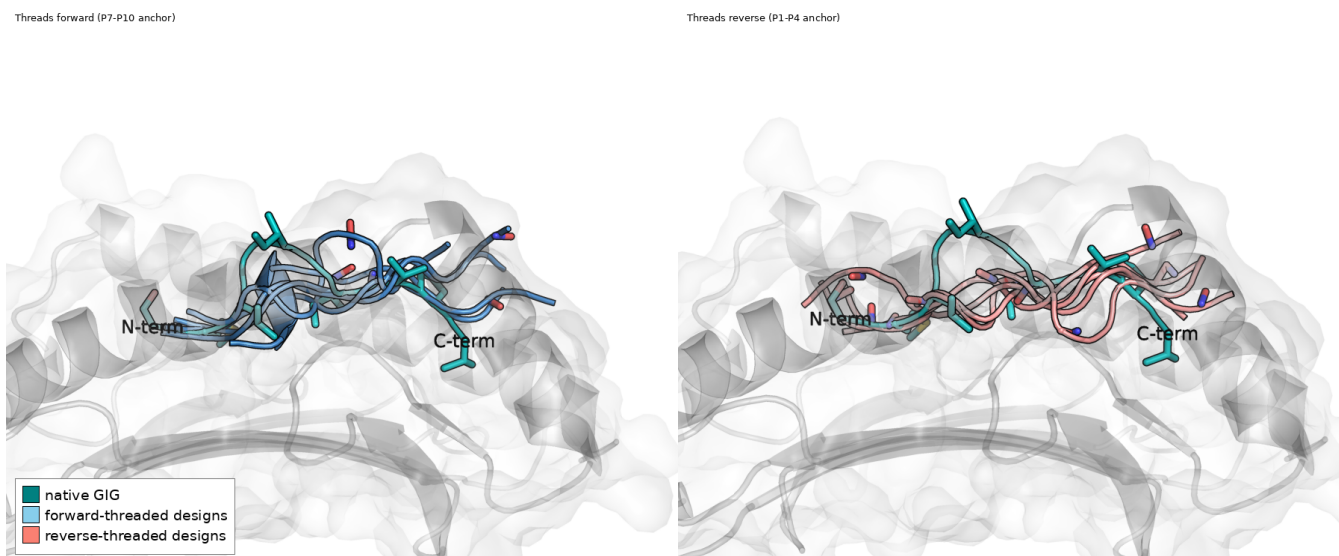


Figure 5: **Forward vs. reverse threading, structural overlay** (C18, 5 representative designs per panel, from a 186-design cohort). Native GIG peptide in teal (N-term/C-term labeled) against the MHC groove (white surface). Left panel: forward-threaded designs (sky blue) run N-to-C the same direction as the native peptide. Right panel: reverse-threaded designs (salmon) anchor the *opposite* end in the same F-pocket — the mirror-image backbone described in §3.4, made directly visible rather than inferred from a distribution.

conditioning	contacts	N	forward	reverse	ambiguous
C18	18	186	93 (50.0%)	87 (46.8%)	6 (3.2%)
C9	9	204	97 (47.5%)	92 (45.1%)	15 (7.4%)

Table 6: Threading classification, two representative conditioning cells.

of population mixing rather than a genuine failure mode. All recovery/crossing statistics in §3.1–3.3 are reported on the forward-threaded subset ($N = 600$) for this reason.

3.5 Templating, not contacts, controls a localized register signal

Sections 3.2–3.4 establish *when* recovery and crossing occur (broad conditioning, §3.2), and what does not further explain their rate (which specific rich scheme is used, §3.3, or backbone threading, §3.4) — but not *why* contact conditioning plateaus at all. To test this directly we ran a separate experiment: instead of conditioning on receptor-side contacts, we fixed an explicit fraction of the native peptide backbone itself as a structural template and let RFdiffusion generate only the remainder. Register recovery scaled sharply with the fraction of geometry supplied this way:

Templating 8/10 residues recovers register to sub-thermal precision ($0.42\text{\AA}$ median, tighter than the native peptide’s own $\sim 0.9\text{\AA}$ thermal envelope at 310K). By contrast, contact-hotspot count alone — up to 42 residues, exceeding the information content of the 8-residue geometric case by residue count — does not break a $\sim 2\text{--}2.5\text{\AA}$ floor, and hotspot count does not correlate with recovery quality (Spearman $\rho \approx 0.17$, not significant; note the `fix0` level of this same ladder, which retains the *full* rich hotspot list but zero tem-

plate, is the worst-performing level at $\sim 11.4\text{\AA}$ median — direct evidence that hotspot conditioning by itself, at its richest, does not recover the target conformation). We interpret this as reflecting distinct mechanisms of action: a contact hotspot biases the predicted position of a *neighboring*, receptor-facing residue, but does not directly constrain a peptide residue’s own backbone dihedral angles — the coordinate on which register is actually encoded.

This localization is direct and measurable: per-residue deviation between the two native conformations, expressed in native-MD thermal-fluctuation units ($z = \text{inter-conformation separation} / \text{native positional RMSF at that position}$), rises from statistical noise at the N-terminus to a pronounced signal at the C-terminal anchor ($z = 0.6$ at P1; 1.3 at P4; 4.9 at P7; 8.1 at P8; 8.5 at P9; 14.5 at P10), with native MD RMSF at these C-terminal positions remaining tight ($\approx 0.37\text{\AA}$), confirming the signal reflects genuine geometric separation rather than amplified noise. This motivates the anchor-identity-based scoring used throughout (§2.3): a whole-peptide-averaged RMSD dilutes this sharp, spatially confined signal across five register-neutral N-terminal positions.

residues templated (of 10)	10	8	6	4	2	0
median C α RMSD to own conformation (Å)	0.07	0.42	0.89	2.06	2.42	~14.1

Table 7: Register recovery vs. fraction of native peptide backbone directly templated.

3.6 Null/background campaign: the completed control admits no accepted design

Sections 3.1–3.5 compare conditioning schemes to one another, but do not by themselves establish how the recovery and crossing rates compare to a fully unconditioned RFdiffusion draw. To measure that background directly, we generated a dedicated campaign of 932 de novo backbones (null0, §2.5) with the receptor held fixed as structural context but **no** `ppi.hotspot_res` and no peptide template.

Scored with the *identical* three-criterion pipeline applied to the main corpus and adjudicated against the same 310K DRG band, **0 of 932** unconditioned draws pass the joint acceptance test (Clopper–Pearson 95% CI 0–0.40%). The more informative observation is *how* they fail. Taken one criterion at a time, the two register-defining gates each admit background: the F-pocket anchor-identity criterion alone is met by 114/932 draws (12.2%), the burial-depth criterion alone by 42/932 (4.5%), and the two together by 10/932 — while the conformational-proximity gate and the full three-criterion conjunction each admit **none**. The null thus supplies an empirical, not merely first-principles (§2.3), justification for the joint test’s stringency: no single register-defining criterion is clean, and only requiring all three jointly yields a zero background — precisely the false-positive failure mode the joint test is built to reject.

The unconditioned draws land well outside the envelope (Fig. 6), and the two scaffolds differ in how far: the 6AM5-context draws never approach the groove (register undefined), while the 6AMU-context draws come closer but remain outside the DRG band. Even the closest unconditioned draw is farther from its native conformation than every accepted de novo design in §3.1. The conclusion this baseline supports is deliberately narrow: in-register groove geometry does not arise as a zero-conditioning background artifact. This baseline is silent on register *selection* (whether conditioning steers the cognate versus the alternate register), which is addressed separately in §3.5.

This completed control (Clopper–Pearson 95% CI for 0/932: 0–0.40%; Fisher’s exact vs. the de novo recovery/crossing rate) confirms that contact conditioning provides real information above a zero-information background. We note that the `fix0` templating-ladder cells (full hotspot list, zero template; §3.5) are *not* a substitute for this measurement, since they still carry the richest hotspot conditioning tested anywhere in this corpus — they are a positive-conditioning control, not a null.

4 Discussion

Three findings jointly define the current capability boundary. First, recovery and crossing of peptide register by contact-conditioned de novo backbone generation are real, structurally validated events, not statistical noise or scoring ar-

tifacts — each reported design passes an independent three-criterion test referenced against thermal motion rather than a static target. Second, achieving either outcome appears to require a minimum breadth of conditioning (coverage of high-contact-degree residues on both the MHC and TCR side), but not a specific curated subset of them — the operative variable is coverage, not composition. Third, once that breadth is present, outcome frequency is governed by how many designs are sampled rather than by which qualifying conditioning is chosen — a result that reframes the practical question from “which contacts should be used” to “how much sampling budget is available,” a comparatively more tractable lever.

The mechanistic account in §3.5 is consistent with this picture: contact hotspots constrain a *neighboring* residue’s predicted position without directly constraining a peptide residue’s own backbone torsions, so increasing contact count broadens the region of conformational space the model is nudged toward without narrowing precision at the specific coordinate (anchor-residue backbone geometry) that defines register. Direct geometric templating, which supplies that coordinate explicitly, achieves sub-thermal precision at 8/10 residues — an upper bound on what the architecture can represent, distinct from what contact-only conditioning can currently retrieve.

The N→C threading artifact (§3.4) — that roughly half of all unconstrained backbones bury the mirror-image terminus in the anchor pocket, invisible to raw C α RMSD — is a genuine, transferable caution for anyone scoring de novo peptide backbones by RMSD alone, independent of the specific recovery/crossing rates reported here. The completed null-baseline experiment (§3.6) establishes that the observed de novo recovery/crossing rate exceeds a true zero-information background (0/932), which the tier comparisons in §3.2–3.3 cannot by themselves resolve.

5 Limitations and scope

- **Backbone (C α) scope only.** No sequence-, side-chain-, or rotamer-level claim is made; RFdiffusion’s output at this stage carries no designed sequence, and any downstream ProteinMPNN sequence-design results are a separate body of work not represented here. This analysis deliberately stops at backbone generation and does not run ProteinMPNN.
- **Calibration convergence.** The 310K acceptance thresholds (§2.4) derive from a single 50ns MD trajectory per register, and the two runs differ in convergence: the DRG (6AMU) run is well sampled (integrated autocorrelation time 1.9ns, effective sample size ≈ 13) but the GIG (6AM5) run is not (≈ 2), which is why redirection is adjudicated only against the DRG band and why acceptance

thresholds are reported with block-bootstrap 95% confidence intervals rather than as point values. We expect further simulation to refine these thresholds rather than overturn the qualitative result, given the margin by which the reported designs clear them, but exact Å cutoffs should be treated as current-best rather than final.

- **Small absolute event counts.** Although the null baseline is now complete (0/932, §3.6), the de novo recovery and crossing counts themselves remain small (7 designs within the DRG band’s 95% CI over 8,722 draws), so the corpus-wide rates carry wide confidence intervals and are reported with them throughout; they should be read as establishing that these events occur above background, not as precise frequency estimates.
- **Overlapping campaign membership.** 16 of the 55 conditioning cells pool draws from more than one campaign launched at different times (§2.2); this is disclosed for transparency and is consistent with, not a threat to, the draw-count correlation reported in §3.3.
- **Controlled breadth/depth sub-experiments.** The conditioning-breadth (§3.2) and sampling-depth (§3.3) dissections are conducted on a separate controlled matched-attempt sub-campaign (equal draws per scheme, to isolate those variables) rather than on the full de novo corpus of §3.1, and rest on small qualifying-design counts within it; the rich-vs-sparse and C18-specific comparisons carry correspondingly wide confidence intervals and should be read as suggestive rather than conclusive.
- **Single system.** All results pertain to one TCR–pMHC pair, one class-I allele, and one force field; no claim is made regarding generalization to other cross-reactive systems, TCRs, or HLA alleles.

6 Data and code availability

Design generation, scoring, and MD analysis scripts are maintained in the project repository (`py/score_denovo_designs.py` for register scoring and threading classification; `py/native_md_components.py` for MD trajectory processing; `jobs/` for RFdiffusion campaign drivers, including `jobs/run_null_baseline.sh` for the §3.6 null/background campaign). Corresponding analysis notebooks (numbered 00–16, grouped by methodology) are indexed in `ANALYSIS_WALKTHROUGH.md`; the MD-calibrated baseline and joint acceptance test underlying §2.3–2.4 and §3.1 are implemented in `01_md_calibrated_baseline.ipynb`.

Supplementary Material

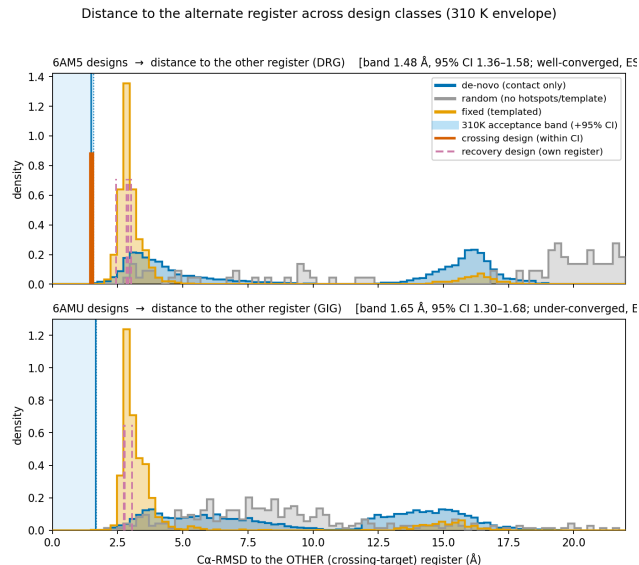


Figure 6: Distribution of $C\alpha$ RMSD to the alternate (crossing-target) register across design classes, per crystal. Density histograms for de novo (contact-conditioned), random (no hotspots, no template), and fixed (templated) backbones. The shaded region marks the 310K native acceptance band with its block-bootstrap 95% confidence interval (§2.4); for 6AM5 designs the target is the well-converged DRG band ($\leq 1.48\text{\AA}$, CI 1.36–1.58), the register into which all observed de novo crossings fall. Vertical lines mark the individual de novo designs judged as crossing (within the DRG band’s 95% CI: 1.44, 1.52, 1.54Å) and the de novo recoveries. Every design class places the overwhelming majority of its mass well beyond the band; only a handful of contact-conditioned designs reach its edge, and the random control places none there. The 6AMU→GIG panel is shown for completeness against the under-sampled GIG band (ESS ≈ 2 , §2.4) and is not used to adjudicate redirection.